

RAGHAV AWASTHI

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EDUCATION

University College London (UCL)

Bachelor of Science in Computer Science, First Class Honours

London, UK

Sep 2023 - Jun 2026

Dubai College

*A*A*A*A*A in A-Levels and 11 grade 9s in GCSEs*

Dubai, UAE

Aug 2016 - Jun 2023

EXPERIENCE

AI/ML Engineering Intern

VotalAI

Jul 2026 - Present

San Francisco, CA, USA

- Shipped Votal, NVIDIA Nemotron and dual-model guard routing across a 24-file feature branch by extending vLLM serving and synchronous/asynchronous fan-out with concurrent verdict aggregation.
- Preserved backward compatibility across 1,066 offline tests by adding 47 routing and launcher tests plus two live contract cases covering payload construction, response parsing and failure handling.
- Hardened the AI gateway against five weakness classes by fixing hCaptcha verification, backend URL handling, subprocess model paths, tenant regexes and LLM tool-call validation for CWE-288, CWE-918, CWE-78, ReDoS and prompt injection.
- Used AI coding tools to accelerate design, implementation, regression-test generation and documentation while manually validating NVIDIA NIM/vLLM contracts and reviewing generated changes before commit.

Cybersecurity Analyst

Barclays

Jun 2025 - Aug 2025

Knutsford, UK

- Projected a 20-30% shorter PQC transition across 30+ services by producing a 60-page PQC/HSM security assessment, capability and gap matrix, and five-phase implementation roadmap.
- Cut certificate triage time 25% and flagged 30+ high-risk certificates by building Python services for CA-chain enrichment, root-cause classification, ServiceNow normalisation, risk scoring and OpenSSL path-frequency analysis over million-row extracts.
- Developed 15+ KQL/SPL threat-hunting queries against live Microsoft Defender, Sentinel and Splunk logs, promoting 7+ detections to production and increasing high-signal alerts approximately 20% in pilot; expanded a signal with 100+ threat terms.
- Personally triaged and investigated 210+ incidents by correlating live security telemetry to distinguish actionable threats from benign activity.
- Completed Copilot prompt-injection testing and presented a GenAI threat assessment to Cyber Operations covering prompt injection, data exfiltration, model poisoning, insecure code generation and mitigations.

Full-Stack Developer

Adamas Data

Aug 2024 - Oct 2024

London, UK

- Kept median query latency below 3 seconds across 140M+ rows and 150+ filter combinations by indexing PostgreSQL and PostGIS queries behind 30+ versioned FastAPI endpoints.
- Enabled rollback in under 10 minutes by containerising and automating CI/CD for a production FastAPI, PostgreSQL and Next.js application delivered through the complete SDLC.

EXPERIENCE (CONTINUED)

AI/ML and Quantum Security Intern

Jun 2024 - Aug 2024

Tech Mahindra

Pune, India

- Raised domain F1 15%, cut mean GPU latency 20% and reduced peak VRAM 60% by leading five engineers to 4-bit LoRA/PEFT fine-tune a 7B model on 120k curated records and benchmark 6k questions.
- Improved retrieval MRR 25% and cut hallucination rate to 7% by converting 295 multilingual PDFs (4k pages) into 12k chunks and 24k QA pairs with Selenium, PyMuPDF/Tesseract, translation, BM25 and MiniLM; limited manual correction to 10%.
- Demonstrated 20 authenticated Flask/Gradio endpoints to stakeholders, achieving 13% English WER, 20% Indonesian WER, 11% Thai CER and approximately 2-second short-request latency across Thai LLM, Whisper/Wav2Vec2 ASR and VITS TTS workflows; kept checkpoint A/B tests under five minutes.
- Completed 100 hybrid TLS 1.3 handshakes at approximately 20 ms overhead by implementing ML-KEM/Kyber512 NTT arithmetic, key generation, encapsulation and decapsulation in Python; validated 800-byte public keys, 768-byte ciphertexts and 32-byte shared secrets while comparing NIST PQC finalists.
- Executed 100-qubit transverse-field Ising circuits on IBM Brisbane with up to 10 Trotter configurations, 12 parameter values and approximately 4,096 shots per circuit by using Qiskit Runtime, SamplerV2/EstimatorV2 and XY4 dynamical decoupling.
- Coordinated four teammates across three to four workstreams, reviewed approximately 12 changes, ran six to eight progress reviews and presented the completed work to approximately 10 stakeholders.

PROJECTS

Walrus-Cache-CDN (UCL Dissertation) | Rust, RocksDB, Tokio, Sui, Prometheus

Jun 2025 - Present

- Increased retrieval throughput 4.2x and cut average latency from 85.0 ms to 20.5 ms on a reproducible 100k-request benchmark by building a lifecycle-aware two-tier Rust cache for Walrus decentralised storage.
- Achieved an 81.7% hit rate by combining W-TinyLFU/LRU admission, persistent RocksDB storage, asynchronous event ingestion and deterministic invalidation by blob ID, tag and epoch with Prometheus observability.

Quantum Proximity Gateway (IBM and UCL) | Python, Rust, C++, Next.js

Oct 2024 - May 2025

- Automated hot-desk PC authentication by integrating ESP32 BLE proximity, Raspberry Pi facial recognition and Pico USB-HID emulation into a distributed multi-factor edge-to-cloud system.

Playo - Audiobook Generator | Python, Firebase, OpenAI, ElevenLabs, TypeScript

Jan 2024

- Delivered end-to-end personalised audiobook generation by combining user preferences with OpenAI story generation, ElevenLabs speech and a TypeScript/Firebase mobile application.

Assembly Code Visualiser | Node.js, Express.js, SQL, Redis

Jun 2022 - Mar 2023

- Made computer architecture easier to learn by building a web assembly IDE and interactive Little Man Computer simulator exposing CPU/RAM state and educator-authored challenges.

Post-Quantum Cryptography Research | Pioneer Academics, C, Linux

Feb 2022 - Jul 2022

- Produced a supervised research paper comparing code-, lattice- and multivariate-based NIST Round 3 PQC finalists through mathematical analysis and empirical C benchmarks.

TECHNICAL SKILLS

Languages: Python, Rust, C/C++, Java, Go, JavaScript/TypeScript, SQL, Bash, KQL, SPL, HTML/CSS

AI/ML: PyTorch, TensorFlow, Hugging Face, vLLM, LoRA/PEFT, LLMs, RAG, BM25, MiniLM, Whisper, Wav2Vec2, VITS, prompt engineering, model evaluation

Backend/Web: FastAPI, Flask, Node.js, React/Next.js, REST APIs, microservices, Gradio, JWT

Data/Storage: RocksDB, PostgreSQL, PostGIS, Redis, SQLite, SQLAlchemy, Pandas, NumPy, Firebase

Security: Threat hunting, detection engineering, MITRE ATT&CK, Defender, Sentinel, Splunk, PKI, HSMs, certificates, PQC, ML-KEM, TLS 1.3, threat modelling, application/API security

Infrastructure: Tokio, Docker, Linux/Unix, GCP, CI/CD, Prometheus, Git, ServiceNow, Selenium, JUnit, Qiskit, automated testing, AI-assisted development

OPEN SOURCE AND LEADERSHIP

- Delivered merged contributions to Microsoft Terminal, mactop and Docusaurus, including a keyboard-driven Go help view and documentation-navigation improvements; worked as a technical writer.
- Served as a UCL programming tutor and cricket vice-captain; previously led the Dubai Computing Community and captained or coached cricket and volleyball teams.